

Unit 3, Lesson 4: Consecutive Angles



Benchmark

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.



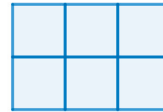
Learning Targets

- ☐ I can determine the relationships of consecutive exterior and interior angles.
- ☐ I can solve problems that involve angles formed by parallel lines intersected by a transversal.

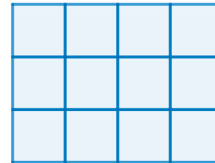


3.4.1 Warm-Up: Squares in Rectangles

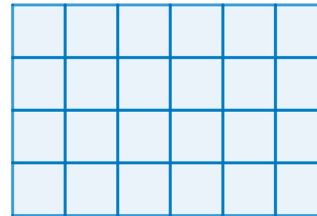
Three rectangles and their descriptions are shown.



A 2×3 rectangle contains 6 unit squares.



A 3×4 rectangle contains 12 unit squares.



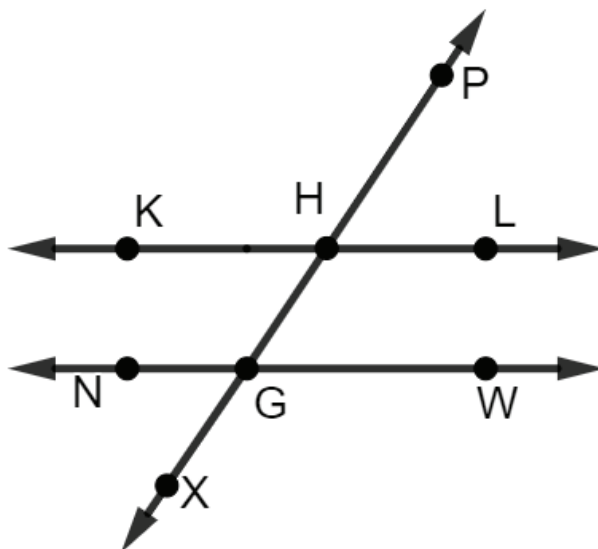
A 4×6 rectangle contains 24 unit squares.

1. What are the dimensions of a rectangle that contains exactly 100 unit squares?
2. List any other size rectangles that contain exactly 100 unit squares.



3.4.2 Collaboration: Consecutive Interior and Exterior Angles

1. $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$ where $\overleftrightarrow{XP}$ is a transversal.



- a. Work with your partner to determine if each pair of angles could be congruent by using corresponding, alternate exterior, alternate interior, or vertical angles.

Angles	Congruent?
$\angle NGH$ and $\angle KHG$	
$\angle PHL$ and $\angle WGX$	

- b. Work with your partner to determine which pair of angles in the table could be described as consecutive interior angles.

- c. Name another pair of consecutive interior angles.
- d. Work with your partner to write a conjecture about the relationship that is true for consecutive interior angles.
- e. Ask another classmate for their conjecture about consecutive interior angles. Record their conjecture and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials

- f. Compare the conjecture from your classmate with the conjecture you and your partner wrote. If necessary, make any adjustments to your conjecture.
- g. $\angle PHL$ and $\angle WGX$ are consecutive exterior angles. Name another pair of consecutive exterior angles.

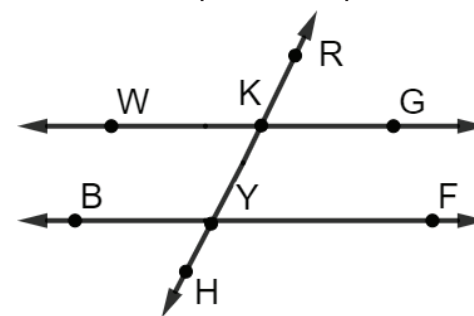
- h. Work with your partner to write a true conjecture about the relationship between consecutive exterior angles.

2. A partially completed proof is shown.

- a. Work with your partner to complete the proof.

Given: $\overleftrightarrow{WG} \parallel \overleftrightarrow{BF}$

Prove: $\angle WKY$ and $\angle BYK$ are supplementary.

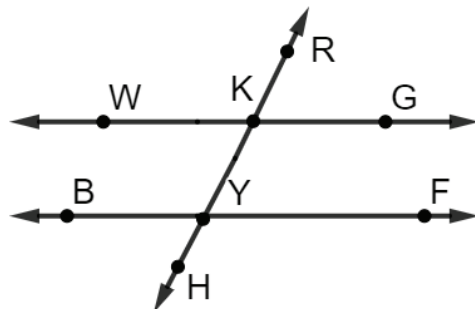


Statement	Reason
1. $\overleftrightarrow{WG} \parallel \overleftrightarrow{BF}$	1. Given
2. $\angle WKY \cong$ _____	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
3. $\angle BYH$ and $\angle BYK$ are a linear pair	3. Definition of a linear pair
4.	4. If two angles form a linear pair, then they are supplementary.
5. $m\angle BYH + m\angle BYK = 180^\circ$	5. Definition of supplementary angles
6. $m\angle WKY =$ _____	6.
7. $m\angle$ _____ $+ m\angle$ _____ $= 180^\circ$	7. Substitution
8. $\angle WKY$ and $\angle BYK$ are supplementary.	8. Definition of supplementary angles

- b. Check your proof with another classmate. If necessary, make any adjustments.

3. Work with your partner to complete the proof.

Given: $\overleftrightarrow{WG} \parallel \overleftrightarrow{BF}$
 Prove: $\angle RKG$ and $\angle FYH$
 are supplementary.



Statement	Reason
1. $\overleftrightarrow{WG} \parallel \overleftrightarrow{BF}$	1. Given
2. $\angle RKG \cong$ _____	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
3. $\angle KYF$ and $\angle FYH$ are a linear pair	3. Definition of a linear pair
4.	4. If two angles form a linear pair, then they are supplementary.
5.	5. Definition of supplementary angles
6.	6.
7. $m\angle$ _____ + $m\angle$ _____ = 180°	7. Substitution
8. $\angle RKG$ and $\angle FYH$ are supplementary.	8. Definition of supplementary angles

4. Mistakes can be an opportunity to learn.

- a. What mistakes did you make in the first proof that helped you in the second proof?

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

- b. If you did not make a mistake in the first proof, describe how the proofs were similar.



Consecutive Interior Angles Theorem

If two parallel lines are intersected by a transversal, then interior angles on the same side of the transversal are supplementary.



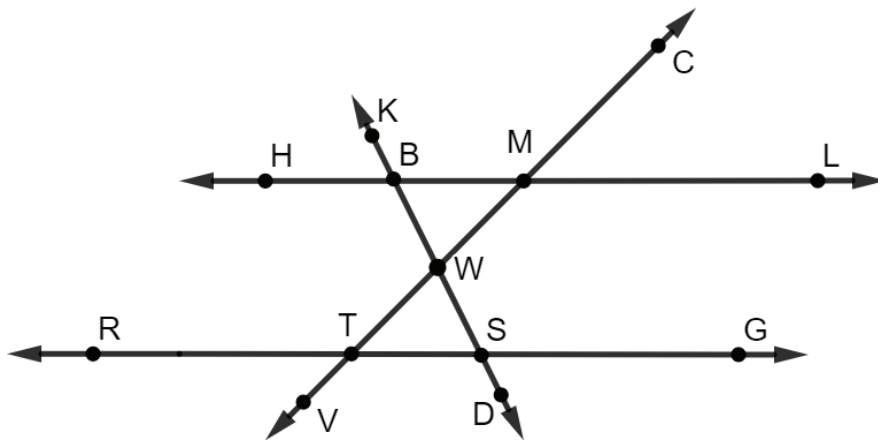
Consecutive Exterior Angles Theorem

If two parallel lines are intersected by a transversal, then exterior angles on the same side of the transversal are supplementary.



3.4.3 Guided Practice: Angles in Parallel Lines

1. $\overleftrightarrow{HL} \parallel \overleftrightarrow{RG}$ where $\overleftrightarrow{CV}$ and $\overleftrightarrow{KD}$ are transversals with $m\angle KBM = 117^\circ$ and $m\angle VTS = 134^\circ$.



Determine the measurement of each angle.

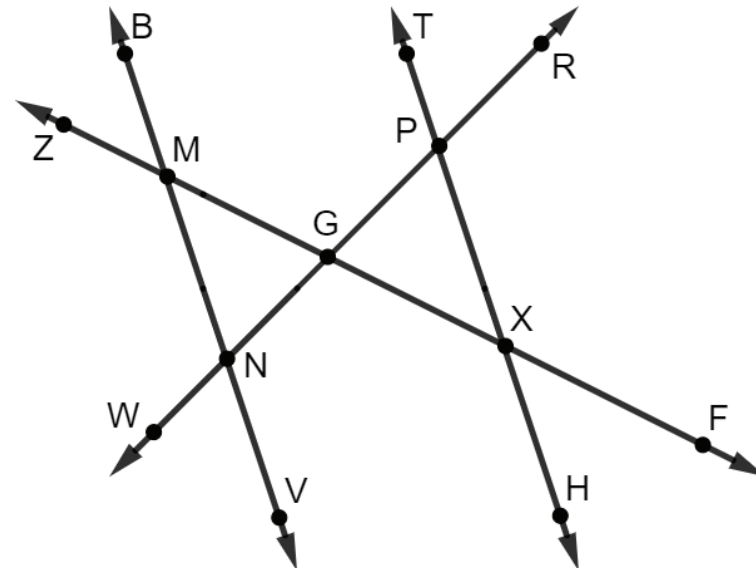
$$m\angle DSG =$$

$$m\angle LMC =$$

$$m\angle RTV =$$

$$m\angle BWM =$$

2. In the figure shown, $\overleftrightarrow{BV} \parallel \overleftrightarrow{TH}$, $m\angle GXH = (5(2x - 1))^\circ$, $m\angle GMN = (4x - 11)^\circ$, $m\angle XPG = (3y - 8)^\circ$, and $m\angle VNG = (7y - 52)^\circ$.



- a. Find the value of x .

- b. Find the value of y .



3.4.4 Wrap-Up: Reflection

1. Rate your confidence in solving problems using these angle pairs.

I can solve number problems involving....	I am an expert and could help others.	I can hold my own.	I am getting there.	I would like more help or practice.
Congruent Supplements				
Congruent Complements				
Vertical Angles				
Consecutive Interior Angles				
Consecutive Exterior Angles				
Alternate Interior Angles				
Alternate Exterior Angles				
Corresponding Angles				



Unit 3, Lesson 5: Converse of the Alternate Interior Angles Theorem



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

MA.912.LT.4.3 Identify and accurately interpret “if...then,” “if and only if,” “all” and “not” statements. Find the converse, inverse, and contrapositive of a statement.



Learning Targets

- ☐ I can write and use the converse of a postulate or theorem.
- ☐ I can use postulates and theorems about parallel lines to determine if lines are parallel.